

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

ELECTRIC POWER INVERTER CONVERTER MODULE (EPICM) - HYBRID ELECTRIC VEHICLE

WARNINGS:

- A risk assessment must be performed before any work is undertaken.
- All hybrid electric vehicle work must be performed by a suitably qualified person.
- Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.
- The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack,

take the following action:

- Evacuate the area
- Notify manager
- Do not breathe smoke/vapour
- Contain spillage using spill kit
- Wash any spillage off body and clothing (remove contaminated clothing)
- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Electric Power Inverter Converter Module (EPICM) - Hybrid Electric Vehicle. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1206-68	Crash Occured - Event information	Event information - The engine control module has received a crash signal from the restraints control module	This DTC is set if the restraints control module has deployed the restraints systems following activation of the crash sensors. Check the restraints control module for DTCs and refer to the relevant DTC index
P0562-17	System Voltage Low - Circuit voltage above threshold	Charging system fault	Using the manufacturer approved diagnostic system, check for charging system related DTCs
P0563-16	System Voltage High - Circuit voltage below threshold	- Electric power inverter converter low voltage power or ground circuit open circuit, high resistance - Battery fault - Charging system fault	- Refer to the electrical circuit diagrams and check the electric power inverter converter low voltage power and ground circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the primary battery - Using the manufacturer approved diagnostic system, check for charging system related DTCs
P0606-49	Control Module Processor - Internal electronic failure	Electric power inverter converter module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0606-96	Control Module Processor - Component internal failure	Electric power inverter converter module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0A05-1C	Motor Electronics Coolant Pump A	Electric power inverter	Refer to the electrical circuit diagrams and check the electric

	Control Circuit / Open - Circuit voltage out of range	converter coolant pump power or ground circuit open circuit, high resistance	power inverter converter coolant pump power and ground circuits for open circuit, high resistance
P0A06-7B	Motor Electronics Coolant Pump A Control Circuit Low - Low fluid level	▪ Cooling system leaking or blocked	▪ Refer to the relevant section of the workshop manual and test the cooling system
P0A07-4B	Motor Electronics Coolant Pump Control Circuit High - Over temperature	▪ Cooling system leaking or blocked ▪ Electric power inverter converter coolant pump internal failure	▪ Refer to the relevant section of the workshop manual and test the cooling system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter coolant pump
P0A0A-13	High Voltage System Interlock Circuit - Circuit open	▪ High voltage connectors not connected correctly ▪ High voltage interlock loop circuit (battery energy control module to electric power inverter converter) open circuit, high resistance	NOTE: The high voltage interlock loop circuit will be open circuit if any high voltage connector is not connected or a connector mechanical latch is not fully engaged ▪ Check that the high voltage connectors are connected correctly ▪ Refer to the electrical circuit diagrams and check the high voltage interlock loop circuit for open circuit, high resistance
P0A1B-95	Drive Motor "A" Control Module - Incorrect assembly	▪ Electric power inverter converter power or ground circuit open circuit, high resistance	NOTE: Rectify any high voltage interlock loop DTCs first

		▪ High voltage circuit to the motor generator open circuit ▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Refer to the electrical circuit diagrams and check the electric power inverter converter power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for open circuit ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A1B-96	Drive Motor "A" Control Module - Component internal failure	▪ Electric power inverter converter internal failure ▪ Software failure ▪ Electric power inverter converter module failure	▪ Using the manufacturer approved diagnostic system, re-configure the electric power inverter converter module with the latest level software. If the fault persists, install a new electric power inverter converter
P0A2B-00	Drive Motor "A" Temperature Sensor Circuit Range/Performance - No sub type information	▪ Motor generator temperature sensor circuit short circuit to ground, short circuit to power, open circuit	▪ Refer to the electrical circuit diagrams and check the motor generator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit
P0A2C-31	Drive Motor "A" Temperature Sensor Circuit Low - No signal	▪ Motor generator temperature sensor circuit short circuit to ground, short circuit to power,	▪ Refer to the electrical circuit diagrams and check the motor generator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit

		open circuit	
P0A2D-31	Drive Motor "A" Temperature Sensor Circuit High - No signal	Motor generator temperature sensor circuit short circuit to ground, short circuit to power, open circuit	Refer to the electrical circuit diagrams and check the motor generator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit
P0A2E-31	Drive Motor "A" Temperature Sensor Circuit Intermittent - No signal	Motor generator temperature sensor circuit short circuit to ground, short circuit to power, open circuit	Refer to the electrical circuit diagrams and check the motor generator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit
P0A2F-4B	Drive Motor "A" Over Temperature - Over temperature	Electric power inverter converter cooling system fault	NOTE: Rectify any motor generator coolant pump DTCs first Refer to the relevant section of the workshop manual and test the electric power inverter converter cooling system
P0A3C-98	Drive Motor "A" Inverter Over Temperature - Component or system over temperature	Electric power inverter converter cooling system fault	NOTE: Rectify any electric power inverter converter coolant pump DTCs first Refer to the relevant section of the workshop manual and test the electric power inverter converter cooling system
P0A3F-13	Drive Motor "A" Position Sensor Circuit - Circuit	Motor generator low voltage circuit short	Refer to the electrical circuit diagrams and check the motor generator low voltage circuits

	open	circuit to ground, short circuit to power, open circuit, high resistance	for short circuit to ground, short circuit to power, open circuit, high resistance
P0A3F-49	Drive Motor "A" Position Sensor Circuit - Internal electronic failure	Motor generator low voltage circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the motor generator low voltage circuits for short circuit to ground, short circuit to power, open circuit, high resistance
P0A40-96	Drive Motor "A" Position Sensor Circuit Range/Performance - Component internal failure	Electric power inverter converter module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0A44-85	Drive Motor "A" Position Sensor Circuit Overspeed - Signal above allowable range	- Automatic transmission internal failure - Motor generator over speed	Using the manufacturer approved diagnostic system, check the transmission control module for related DTCs and refer to the relevant DTC index
P0A5D-13	Drive Motor "A" Phase U Current - Circuit open	- High voltage circuit to the motor generator open circuit - Electric power inverter converter internal failure - Motor generator internal failure	NOTE: Rectify any high voltage interlock loop DTCs first - Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for open circuit - Disconnect connector C1YB02B from the motor generator and connector C1YB05A from the inverter - Check the resistance between the motor generator connector C1YB05A-2 and the inverter

connector C1YB02B-1. The reading should be close to 0 ohms (retain an image of the reading on the multimeter). If resistance readings are high, or open circuit, repair or replace harness as required

- Check the resistance between the motor generator connector C1YB05A-1 and the inverter connector C1YB02B-2. The reading should be close to 0 ohms (retain an image of the reading on the multimeter). If resistance readings are high, or open circuit, repair or replace harness as required
- Check the resistance between the connector motor generator connector C1YB05A-3 and the inverter connector C1YB02B-3. The reading should be close to 0 ohms (retain an image of the reading on the multimeter). If resistance readings are high, or open circuit, repair or replace harness as required
- Check the resistance between the connector motor generator connector C1YB05A-1 and the motor generator connector C1YB05A-2. The reading should be open circuit (retain an image of the reading on the multimeter). If a short circuit is detected, repair or replace harness as required
- Check the resistance between the connector motor generator connector C1YB05A-1 and the motor generator connector C1YB05A-3. The reading should be open circuit (retain an image of the

			reading on the multimeter). If a short circuit is detected, repair or replace harness as required ▪ Check the resistance between the connector motor generator connector C1YB05A-2 and the motor generator connector C1YB05A-3. The reading should be open circuit (retain an image of the reading on the multimeter). If a short circuit is detected, repair or replace harness as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A5E-49	Drive Motor "A" Phase U Current Low - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A5F-49	Drive Motor "A" Phase U Current High - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

			fault persists, install a new motor generator
P0A60-13	Drive Motor "A" Phase V Current - Circuit open	▪ High voltage circuit to the motor generator open circuit ▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	NOTE: Rectify any high voltage interlock loop DTCs first ▪ Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for open circuit ▪ Disconnect connector C1YB02B from the motor generator and connector C1YB05A from the inverter ▪ Check the resistance between the motor generator connector C1YB05A-2 and the inverter connector C1YB02B-1. The reading should be close to 0 ohms (retain an image of the reading on the multimeter). If resistance readings are high, or open circuit, repair or replace harness as required ▪ Check the resistance between the motor generator connector C1YB05A-1 and the inverter connector C1YB02B-2. The reading should be close to 0 ohms (retain an image of the reading on the multimeter). If resistance readings are high, or open circuit, repair or replace harness as required ▪ Check the resistance between the connector motor generator connector C1YB05A-3 and the inverter connector C1YB02B-3. The reading should be close to 0 ohms (retain an image of the reading on the multimeter). If resistance

readings are high, or open circuit, repair or replace harness as required

- Check the resistance between the connector motor generator connector C1YB05A-1 and the motor generator connector C1YB05A-2. The reading should be open circuit (retain an image of the reading on the multimeter). If a short circuit is detected, repair or replace harness as required
- Check the resistance between the connector motor generator connector C1YB05A-1 and the motor generator connector C1YB05A-3. The reading should be open circuit (retain an image of the reading on the multimeter). If a short circuit is detected, repair or replace harness as required
- Check the resistance between the connector motor generator connector C1YB05A-2 and the motor generator connector C1YB05A-3. The reading should be open circuit (retain an image of the reading on the multimeter). If a short circuit is detected, repair or replace harness as required
- Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
- Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator

P0A61-49	Drive Motor "A" Phase V Current Low - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A62-49	Drive Motor "A" Phase V Current High - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A63-13	Drive Motor "A" Phase W Current - Circuit open	▪ High voltage circuit to the motor generator open circuit ▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	NOTE: Rectify any high voltage interlock loop DTCs first ▪ Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for open circuit ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator

P0A64-49	Drive Motor "A" Phase W Current Low - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A65-49	Drive Motor "A" Phase W Current High - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0A78-49	Drive Motor "A" Inverter Performance - Internal electronic failure	▪ Hybrid electric vehicle battery pack fault ▪ Electric power inverter converter power or ground circuit open circuit, high resistance ▪ Electric power inverter converter internal failure ▪ Software failure ▪ Inverter failure	NOTE: Rectify any high voltage interlock loop DTCs first ▪ Using the manufacturer approved diagnostic system, check the battery energy control module for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the electric power inverter converter power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, re-configure the electric power inverter converter module with the latest level software. If the fault persists, install a new electric power inverter

			converter
P0A94-01	DC/DC Converter Performance - General electrical failure	- Hybrid electric vehicle battery pack fault - Electric power inverter converter power or ground circuit open circuit, high resistance - Electric power inverter converter internal failure - Software failure - DC to DC converter failure	NOTE: Rectify any high voltage interlock loop DTCs first - Using the manufacturer approved diagnostic system, check the battery energy control module for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the electric power inverter converter power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, re-configure the electric power inverter converter module with the latest level software. If the fault persists, install a new electric power inverter converter
P0A94-16	DC/DC Converter Performance - Circuit voltage below threshold	Electric power inverter converter low voltage power or ground circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the electric power inverter converter low voltage power and ground circuits for open circuit, high resistance
P0A94-17	DC/DC Converter Performance - Circuit voltage above threshold	Electric power inverter converter low voltage power or ground circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the electric power inverter converter low voltage power and ground circuits for open circuit, high resistance
P0A94-49	DC/DC Converter Performance - Internal electronic	Electric power inverter converter	Using the manufacturer approved diagnostic system, re-configure the electric power

	failure	internal failure - Software failure - DC to DC converter failure	inverter converter module with the latest level software. If the fault persists, install a new electric power inverter converter
P0AEE-00	Drive Motor Inverter Temperature Sensor A Circuit Range/Performance - No sub type information	- Electric power inverter converter internal failure - Software failure	Using the manufacturer approved diagnostic system, re-configure the electric power inverter converter module with the latest level software. If the fault persists, install a new electric power inverter converter
P0AF1-00	Drive Motor Inverter Temperature Sensor A Circuit Intermittent/Erratic - No sub type information	- Electric power inverter converter power or ground circuit open circuit, high resistance - Electric power inverter converter internal failure	- Refer to the electrical circuit diagrams and check the electric power inverter converter power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0AFA-16	Hybrid Battery System Voltage Low - Circuit voltage below threshold	High voltage circuit to the hybrid electric vehicle battery pack open circuit	Refer to the electrical circuit diagrams and check the high voltage circuit to the hybrid electric vehicle battery pack for open circuit
P0AFB-17	Hybrid Battery System Voltage High - Circuit voltage above threshold	High voltage circuit to the hybrid electric vehicle battery pack open circuit	Refer to the electrical circuit diagrams and check the high voltage circuit to the hybrid electric vehicle battery pack for open circuit
P0B38-1C	Motor Electronics Coolant Pump B Control Circuit / Open - Circuit voltage out of	Motor generator coolant pump power or ground circuit open circuit,	Refer to the electrical circuit diagrams and check the motor generator coolant pump power and ground circuits for open circuit, high resistance

	range	high resistance	
P0B39-7B	Motor Electronics Coolant Pump B Control Circuit Low - Low fluid level	▪ Cooling system leaking or blocked	▪ Refer to the relevant section of the workshop manual and test the cooling system
P0B3A-4B	Motor Electronics Coolant Pump B Control Circuit High - Over temperature	▪ Cooling system leaking or blocked ▪ Motor generator coolant pump internal failure	▪ Refer to the relevant section of the workshop manual and test the cooling system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator coolant pump
P0BE5-1D	Drive Motor "A" Phase U Current Sensor Circuit - Circuit current out of range	▪ High voltage circuit to the motor generator short circuit ▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for short circuit ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0BE6-49	Drive Motor A Phase U Current Sensor Circuit Range/Performance - Internal electronic failure	▪ Electric power inverter converter module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0BE9-1D	Drive Motor A Phase V Current Sensor Circuit - Circuit current out of range	▪ High voltage circuit to the motor generator short circuit ▪ Electric power inverter converter internal failure	▪ Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for short circuit ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

		▪ Motor generator internal failure	electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0BEA-49	Drive Motor A Phase V Current Sensor Circuit Range/Performance - Internal electronic failure	▪ Electric power inverter converter module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0BED-1D	Drive Motor A Phase W Current Sensor Circuit - Circuit current out of range	▪ High voltage circuit to the motor generator short circuit ▪ Electric power inverter converter internal failure ▪ Motor generator internal failure	▪ Refer to the electrical circuit diagrams and check the high voltage circuit to the motor generator for short circuit ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new motor generator
P0BEE-49	Drive Motor A Phase W Current Sensor Circuit Range/Performance - Internal electronic failure	▪ Electric power inverter converter module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter
P0BFD-49	Drive Motor A Phase U-V-W Current Sensor Correlation - Internal electronic failure	▪ Electric power inverter converter internal failure ▪ Software failure	▪ Using the manufacturer approved diagnostic system, re-configure the electric power inverter converter module with the latest level software. If the fault persists, install a new electric power inverter converter

P0C50-13	Drive Motor A Position Sensor Circuit A - Circuit open	Motor generator low voltage circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the motor generator low voltage circuits for short circuit to ground, short circuit to power, open circuit, high resistance
P0C5A-13	Drive Motor B Position Sensor Circuit A - Circuit open	Motor generator low voltage circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the motor generator low voltage circuits for short circuit to ground, short circuit to power, open circuit, high resistance
P0C73-82	Motor Electronics Coolant Pump A Control Performance - Alive / sequence counter incorrect / not updated	Electric power inverter converter coolant pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the electric power inverter converter coolant pump control circuit for short circuit to ground, short circuit to power, open circuit, high resistance
P0C73-93	Motor Electronics Coolant Pump A Control Performance - No operation	Electric power inverter converter coolant pump internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter coolant pump
P0C74-82	Motor Electronics Coolant Pump B Control Performance - Alive / sequence counter incorrect / not updated	Motor generator coolant pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the motor generator coolant pump control circuit for short circuit to ground, short circuit to power, open circuit, high resistance
P0C74-93	Motor Electronics Coolant Pump B Control Performance - No	Motor generator coolant pump internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

	operation		motor generator coolant pump
P0D32-98	DC/DC Converter Over Temperature - Component or system over temperature	▪ Cooling system leaking or blocked ▪ Electric power inverter converter coolant pump internal failure	▪ Refer to the relevant section of the workshop manual and test the cooling system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric power inverter converter coolant pump
P0E4F-03	Motor Electronics Coolant Pump "A" Control Circuit Intermittent/Erratic - FM (Frequency Modulated) / PWM (Pulse Width Modulated) failures	▪ Electric power inverter converter coolant pump control circuit short circuit to ground, short circuit to power	▪ Refer to the electrical circuit diagrams and check the electric power inverter converter coolant pump control circuit for short circuit to ground, short circuit to power
P0E50-03	Motor Electronics Coolant Pump "B" Control Circuit Intermittent/Erratic - FM (Frequency Modulated) / PWM (Pulse Width Modulated) failures	▪ Motor generator coolant pump control circuit short circuit to ground, short circuit to power	▪ Refer to the electrical circuit diagrams and check the motor generator coolant pump control circuit for short circuit to ground, short circuit to power
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	▪ Invalid data received from another control module via the high speed CAN bus (hybrid)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation incorrect	▪ Invalid data received from another control module via the high speed CAN bus (hybrid)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index

U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (hybrid) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0001-96	High Speed CAN Communication Bus - Component internal failure	High speed CAN bus (analogue) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (analogue) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0100-87	Lost Communication With ECM/PCM "A" - Missing message	- Engine control module power or ground circuit open circuit, high resistance - High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Engine system fault	- Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (hybrid) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0146-87	Lost Communication With Gateway "A" - Missing message	Body control module/gateway module assembly power or ground circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the body control module/gateway module assembly power and ground circuits for open circuit, high resistance

		▪ High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Body control module/gateway module assembly system fault	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (hybrid) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the body control module/gateway module assembly for related DTCs and refer to the relevant DTC index
U0167-00	Lost Communication With Vehicle Immobilizer Control Module - No sub type information	▪ Electric steering column lock control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock system fault	▪ Refer to the electrical circuit diagrams and check the electric steering column lock control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the electric steering column lock control module for related DTCs and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Car configuration	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer

		file mismatch with vehicle specification Incorrect electric power inverter converter module installed	approved diagnostic system, check and update the car configuration file as necessary Install a new electric power inverter converter module as necessary
U0426-00	Invalid Data Received From Vehicle Immobilizer Control Module - No sub type information	- Missing/invalid data from the electric steering column lock control module - Electric power inverter converter module is not configured correctly	- Using the manufacturer approved diagnostic system, check the electric steering column lock control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, re-configure the electric power inverter converter module with the latest level software
U3008-79	Control Module Ground "A" - Mechanical linkage failure	Ground sense circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the ground sense circuit for open circuit, high resistance